

DriveWise AI+: Data-Driven Decision Support for Used Car Dealerships

Business Analytics Individual Case Study Report

Dataset: 10,468 cleaned CarDekho listings across 295 cities in India

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1. Problem Statement and Objectives

1.1 Problem Statement

Used car dealerships have to make several decisions at the same time. They need to decide how much a vehicle is worth, whether it is safe to purchase, how quickly it may sell, and whether it should be added to the available inventory. When these decisions are made mainly from asking prices or personal judgement, a dealer may pay too much for a vehicle or keep a slow-moving vehicle for too long. This increases holding cost and reduces the money available for other vehicles. The problem studied in this case is therefore how business analytics can help a used car dealer select vehicles that have reasonable value, lower purchase risk, good expected profit, and faster inventory turnover.

1.2 Objectives

- Build and evaluate machine learning models to estimate the fair market value of used cars.
- Create a combined decision score that considers profit, demand, holding cost, return on investment, depreciation, and purchase risk.
- Use the analytical results to select a vehicle portfolio under a fixed capital budget and lot-capacity constraint.

2. Data Collection and Dataset Description

2.1 Data Collection

The primary dataset was collected by web scraping publicly accessible used-car listings from CarDekho India. A Python crawler was used for collection. The project documentation records that robots.txt was checked before requests, random delays of 2–4 seconds were used between requests, and an academic User-Agent was supplied. The scraper started from 18 major Indian cities and the collected listings covered 295 cities after pagination and location information were processed.

The project documentation reports 10,505 raw listings and 10,468 cleaned and validated listings, covering 28 brands and 295 cities.

Dataset detail	Value
Raw listings	10,505
Cleaned listings	10,468
Brands	28
Cities	295
Main source	CarDekho India
Main target	Listed vehicle price (INR)

2.2 Important Variables

Variable	Meaning	Role
price_inr	Listed price in Indian Rupees	Target variable for price prediction
brand_model	Vehicle manufacturer and model	Important categorical price factors
year_car_age	Registration year and derived age	Measures vehicle age
mileage_km	Odometer mileage	Proxy for vehicle usage
transmission	Manual / Automatic	Used to study price differences
fuel_type	Petrol, Diesel, CNG, Electric, Hybrid	Vehicle powertrain type
city	Seller location	Used for geographic analysis
mileage_per_year	Derived annual mileage	Helps identify unusually high usage
price_vs_model_median_pct	Difference from model median	Used in purchase-risk analysis

3. Data Preparation and Exploratory Analysis

3.1 Data Preparation

1. Duplicate listings were removed using brand, model, year, price, mileage, and city.
2. Brand names were standardized so that the same manufacturer was not treated as different entities.
3. Unrealistic prices, mileages, and registration years were removed using predefined bounds. Only 37 of the 10,505 raw rows were removed at this stage.
4. Categorical variables with fewer levels were one-hot encoded. High-cardinality variables such as model and city were frequency encoded.
5. Additional business variables such as car age, annual mileage, acquisition cost, holding cost, days on market, and condition grade were prepared for the decision-support layer.

3.2 Exploratory Analysis

- The mean listed price was about ₹11.64 lakh, while the median was ₹7.49 lakh. The difference shows that a smaller number of expensive vehicles pull the average upward.
- The average vehicle age was 5.8 years and the average mileage was about 58,410 km. Average annual usage was about 11,240 km.
- Automatic cars had a higher median listed price than manual cars in the collected listings: ₹14.50 lakh compared with ₹5.75 lakh. This is a descriptive difference in this dataset, not a causal conclusion.
- The listings covered 295 cities. Large metropolitan areas had more listings and a larger presence of higher-priced sedans and SUVs, while several Tier-2 locations had more compact cars.

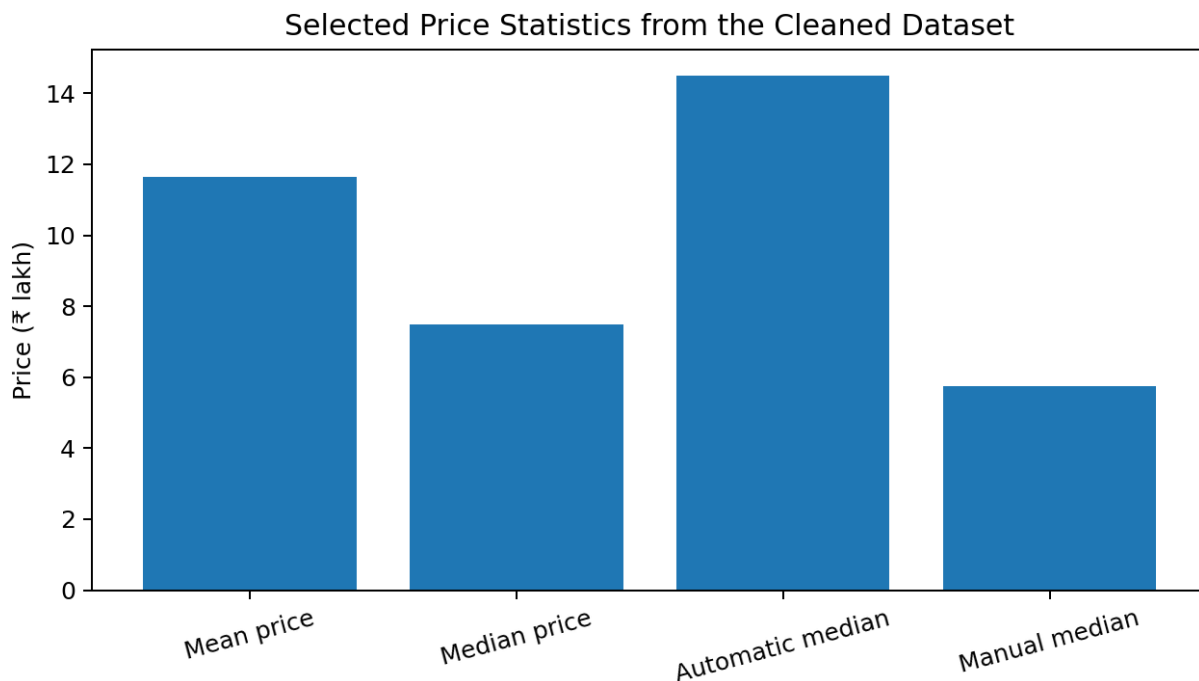


Figure 1. Selected price statistics reported from the cleaned dataset.

4. Analytics Method and Implementation

The case study uses a combination of predictive, descriptive, and prescriptive analytics. The methods were selected because the business problem involves both estimating vehicle value and deciding which vehicles should be purchased.

Method	Purpose
Multiple Linear Regression	Baseline model used to check how well a simple linear relationship explains price.
Random Forest Regressor	Captures non-linear relationships between vehicle attributes and price using many decision trees.
XGBoost Regressor	Gradient-boosting model used for the main fair-value prediction task.
Exponential depreciation model	Used to estimate how vehicle value changes with age for major brands.
Isolation Forest	Unsupervised method used to flag unusual listings that may have higher purchase risk.
Dealer Intelligence Index (DII)	Combines six business factors into a 0–100 score for vehicle-level decision support.
0/1 MILP knapsack model	Selects a portfolio of vehicles while respecting a capital budget and maximum lot capacity.

4.1 Predictive Valuation

The cleaned dataset was divided into 80% training data and 20% test data. This produced 2,094 test listings. Three regression models were compared.

Model	R ²	MAPE	Use in project
Linear Regression	0.5261	67.30%	Baseline
Random Forest	0.8697	20.35%	Non-linear ensemble
XGBoost	0.9129	17.20%	Selected fair-value model

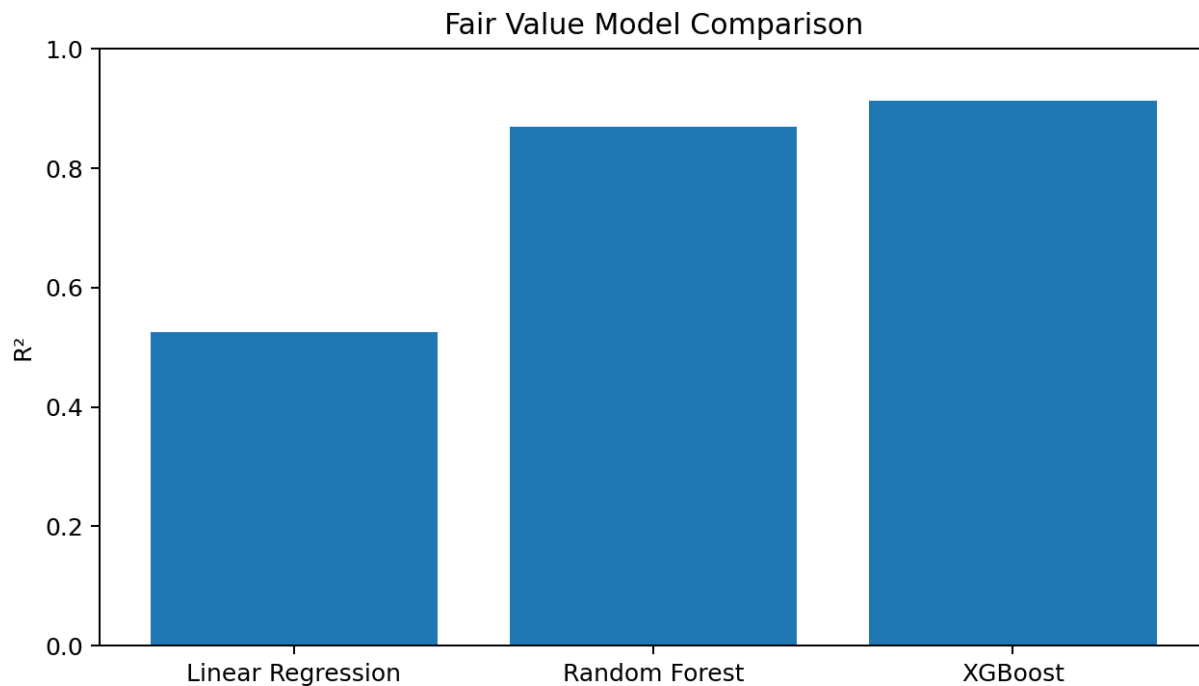


Figure 2. R² comparison of the three fair-value models.

XGBoost gave the highest R^2 among the three models in this experiment. Its test results were $R^2 = 0.9129$, $MAE = ₹1,89,796$ and $MAPE = 17.20\%$. The model was therefore used as the main fair-value model in the later decision-support stages.

4.2 Depreciation and Purchase Risk

Exponential depreciation curves were fitted for major brands to describe the relationship between vehicle age and value. An Isolation Forest model was also trained using variables such as price deviation from the model median, annual mileage, vehicle age, and capital-per-kilometre ratio. Its output was converted to a 0–100 risk score.

4.3 Dealer Intelligence Index

The Dealer Intelligence Index (DII) combines six normalized components: expected profit, demand/turnover, holding-cost efficiency, return on investment, depreciation/value retention, and certainty (inverse anomaly risk). The weights used were 0.30, 0.15, 0.10, 0.20, 0.10, and 0.15 respectively. The purpose is not to replace the individual measures, but to provide one easy-to-read score when a dealer has to compare many vehicles.

4.4 Portfolio Optimization

The portfolio problem was written as a 0/1 knapsack Mixed-Integer Linear Program. Each candidate vehicle is either selected or not selected. The objective is to maximize total expected profit. The main constraints are the available acquisition budget and a maximum lot capacity of 40 vehicles. For the ₹1 crore scenario, the model selected 32 vehicles and reported an expected net profit of ₹40.43 lakh with 40.44% ROI and average DOM of 19.3 days.

5. Comparison with State-of-the-Art Methods

The following comparison uses published studies related to used-car price prediction. The reported scores should not be compared as if they were produced under the same experimental conditions, because the datasets, features, validation procedures, and target definitions are different.

Published study / year	Dataset	Method used	Evaluation metric	Key result	Comparison with this work
Gegic et al. (2019)	1,105 scraped used-car listings from Bosnia and Herzegovina; 797 after processing	ANN, SVM, Random Forest; ensemble classification	Test accuracy	87.38% accuracy for the final ensemble	Both studies use web-collected used-car data and compare ML methods. This work uses 10,468 Indian listings and treats price prediction as regression, then adds risk and portfolio decisions.
Huang et al. (2022)	30,000 Chinese used-car transactions	XGBoost, CatBoost, LightGBM, ANN; model fusion	R^2 , other prediction measures; leave-one-out validation	Four-model fusion reported $R^2 = 0.9845$	The study focuses on valuation accuracy. This work also evaluates valuation, but extends the analysis to purchase risk, business scoring and inventory optimization.
Cui et al. (2022)	Used-car data from a Chinese large-data setting	Iterative XGBoost + LightGBM with deep residual features	Prediction accuracy / comparison with mainstream models	Iterative framework improved prediction performance over individual baselines	Both use tree-based boosting for valuation. This work uses XGBoost as one layer in a wider decision-support pipeline rather than building only a price model.
Nuzulia et al. (2024)	Listings scraped from OLX Indonesia	XGBoost and Random Forest	R^2 and test-time measures	Random Forest reported $R^2 = 0.9485$ under the stated configuration	This work also compares Random Forest and XGBoost, but uses Indian multi-city CarDekho data and adds a business optimization layer.

5.1 Discussion

The published studies show that machine learning is widely used for used-car valuation, especially tree-based models and model combinations. Some studies use much larger transaction datasets, while others use scraped online listings. The main difference in this case study is that price prediction is treated as only one part of the business problem. The project

adds purchase-risk detection, a multi-factor decision score, and portfolio optimization under a capital limit. This makes the output more directly connected to the dealership decision described in the problem statement.

6. Results, Business Insights and Recommendations

6.1 Main Results

- XGBoost achieved $R^2 = 0.9129$, $MAE = ₹1,89,796$ and $MAPE = 17.20\%$ on the reported 2,094-listing test set.
- The top DII decile had average expected profit of ₹2,62,577, 33.2% ROI and 24.6 days on market. The bottom decile had ₹39,236 profit, 6.3% ROI and 78.3 days on market.
- Under a ₹1 crore budget, the MILP portfolio contained 32 vehicles, with reported expected profit of ₹40.43 lakh, 40.44% ROI and 19.3 days average DOM.
- Under the project's severe downside scenario, the DriveWise portfolio retained ₹14.24 lakh of reported net profit, while the Cheapest-First strategy produced a reported loss of ₹1.23 lakh. These are scenario results based on the project's assumptions.

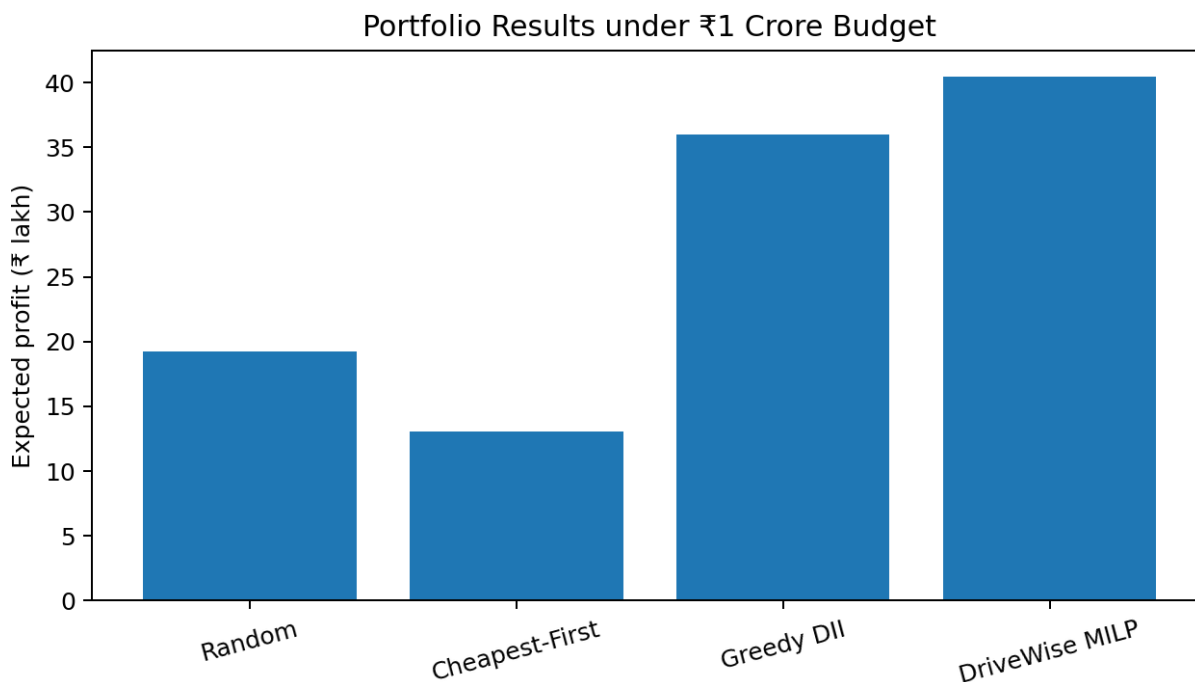


Figure 3. Reported portfolio results under a ₹1 crore budget.

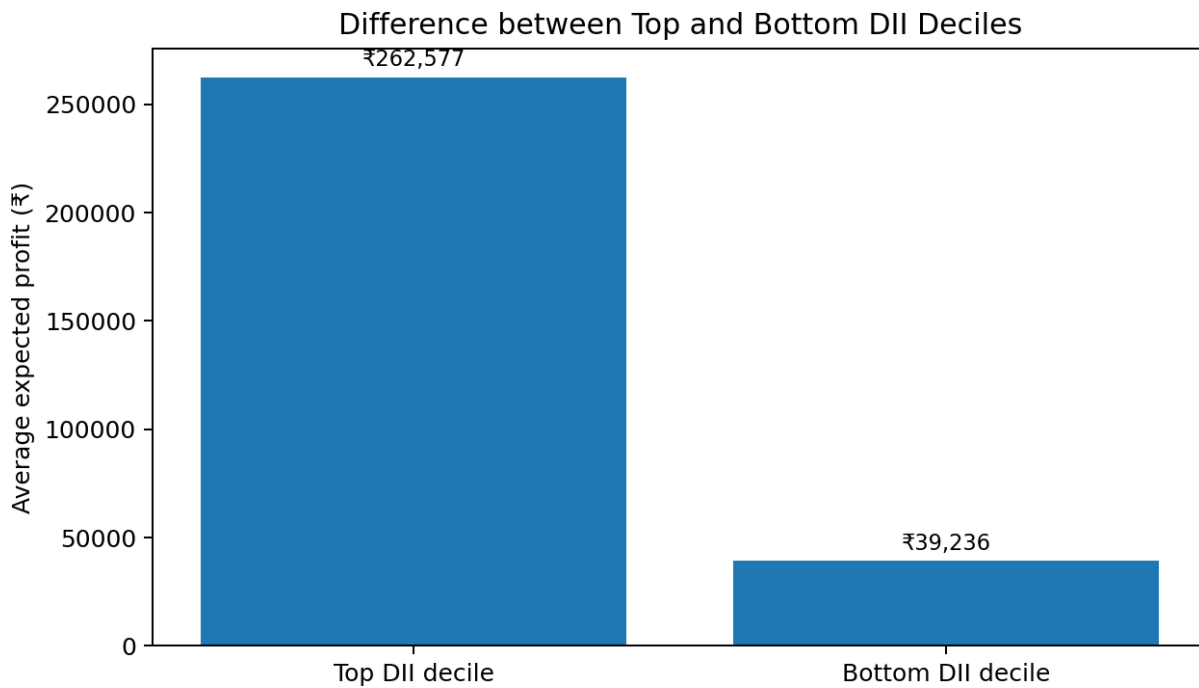


Figure 4. Reported profit difference between the top and bottom DII deciles.

6.2 Business Insights

- Vehicle price alone is not enough for a dealership decision. A vehicle with a good predicted price can still be less attractive if it has high holding cost or slow expected turnover.
- The DII gives managers a simple way to compare several business factors at the same time.
- Portfolio optimization is useful when capital is limited because selecting a set of vehicles is different from selecting the highest-profit vehicle individually.
- Explainable outputs can help managers understand why a vehicle receives a particular score before making a purchase decision.

6.3 Recommendations

- Use predicted fair value as a reference during vehicle acquisition instead of relying only on the seller's asking price.
- Check the purchase-risk score together with expected profit. A high expected margin should not be considered alone if the vehicle has unusual price, age, or usage characteristics.
- Use a portfolio approach when capital is limited. The optimization model can balance several vehicles instead of selecting cars one by one.
- Monitor days on market and holding cost regularly. The project used 0.22% of acquisition capital per day as the carrying-cost assumption and tested dynamic price adjustments of up to $\pm 8\%$.
- Use explainable decision information when presenting recommendations to a dealer. Showing the main factors behind a score can make the analytical output easier to understand.

6.4 Limitations

- The dataset contains online asking/listing prices rather than completed transaction prices. Actual selling prices may therefore differ.
- Days on market, acquisition cost, condition grade and some operational variables are derived or simulated business-layer variables rather than directly scraped fields.
- The model is evaluated on this dataset and should be revalidated when market conditions, regions, or vehicle types change.

- The portfolio results depend on assumptions such as holding cost, resale price adjustment, acquisition discount, and budget. These assumptions should be updated with real dealership data before operational use.

7. Conclusion and References

7.1 Conclusion

This case study shows how Business Analytics can be used beyond simple price prediction. A web-scraped dataset of 10,468 used-car listings was cleaned and analyzed, and several predictive models were compared. XGBoost gave the strongest reported test performance among the three valuation models used in the project. The study then combined valuation with anomaly detection, a multi-factor Dealer Intelligence Index, and a portfolio optimization model. Under the ₹1 crore example budget, the optimization model selected 32 vehicles with a reported expected profit of ₹40.43 lakh. Overall, the project demonstrates a complete path from data collection to analysis and business decision support. The results should be treated as analytical support rather than a replacement for dealer inspection, negotiation, and real transaction data.

7.2 References

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